GATMACH YUOL NYUON

Computer Science Graduate · AI & Machine Learning · Software Development · Data Science

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PROFESSIONAL SUMMARY

Computer Science graduate from JKUAT with deep technical expertise in AI, machine learning, data science, and full-stack software development — and a lived commitment to digital inclusion. Raised and educated in Kakuma Refugee Camp, with direct experience teaching and mentoring refugee youth in technology and academic subjects. Certified by NVIDIA in LLM application development and prompt engineering. Proven ability to design and deliver digital skills instruction, mentor underserved learners, and bridge the gap between complex technology and real-world community impact. Passionate about leveraging technology to create economic opportunity for refugees, women, and marginalised youth.

EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT) *2022 – 2026*
B.Sc. Computer Science · Graduation: August 2026

- Key coursework: Machine Learning, AI, Data Science, Software Engineering, Database Systems, Data Structures & Algorithms etc.

Kakuma Refugee Secondary School *2017 – 2020*
KCSE · Mean Grade: B Plain, 62 Points

Ngundeng Refugee Camp Primary School *2009 – 2016*
KCPE · Score: 357 Marks

PROFESSIONAL EXPERIENCE

Mathematics & Chemistry Teacher *Feb 2022 – Aug 2022*
Windle International Kenya (WIK) — Vision Secondary School, Kakuma Refugee Camp

- Delivered Mathematics and Chemistry instruction to secondary school students in a humanitarian education setting, designing inquiry-based lesson plans that measurably improved student engagement and academic outcomes.
- Provided targeted, inclusive support for learners with special educational needs, adapting teaching materials and methods to ensure equitable learning experiences for all students.
- Collaborated with parents, counsellors, and multi-disciplinary staff to provide holistic learner support, contributing to improved student attendance and retention.
- Operated effectively within the resource constraints typical of refugee camp education environments, demonstrating adaptability and problem-solving under pressure.

Volunteer Mathematics & CRE Teacher *Jan 2021 – Feb 2022*
Kakuma Refugee Secondary School — Kakuma Refugee Camp

- Provided voluntary academic instruction and peer mentorship in a high-need school environment, championing inclusive teaching and equitable access to quality education.
- Contributed to improved academic performance and school culture through sustained community-oriented engagement and curriculum support.

LEADERSHIP & COMMUNITY ENGAGEMENT

Secretary General *Jan 2021 – Sep 2024*
Maikiir Youth Association — Kakuma

- Led executive administration, strategic planning, and governance for a community youth organisation over 3.5 years, developing strong organisational leadership and cross-team coordination skills.
- Designed and delivered community development programmes and youth initiatives with measurable positive social impact within the Kakuma refugee community.
- Managed communications, documentation, and partnerships across a multicultural membership, building strong interpersonal and stakeholder engagement capabilities.

TECHNOLOGY PROJECTS

Farm2Fork — Decentralised Tomato Supply Chain Traceability using Blockchain *2025 – Present*
Final Year Project · Blockchain & Agricultural Supply Chain
Stack: *React Native Expo TypeScript Solidity Ethereum Laravel PHP Smart Contracts IPFS Pinata Axios AsyncStorage SQLite QR Code GPS React Navigation Sepolia Testnet*

- Building a mobile DApp on Expo that uses blockchain smart contracts to deliver end-to-end traceability of tomatoes supply chains — from farm harvest to final consumer — combating food fraud and enabling real-time quality assurance.

Crop Disease Classification System *2024*
Academic Project
Stack: *TensorFlow · CNNs · Python · FastAPI · Docker*

- Built and deployed a deep learning model for real-time crop disease detection via REST API — practical AI applied directly to agricultural challenges affecting smallholder farmers.

Infectious Disease Modeling (Meningitis) *2024*
Academic Project
Stack: *Python · Starsim · Agent-Based Modeling · Statistical Analysis*

- Modelled meningitis transmission dynamics and evaluated vaccination intervention scenarios — applying data science to public health challenges in communities like Kakuma.

Smart Waste Management System *2024*
Academic Project
Stack: *Python · IoT · Data Analytics · Route Optimisation*

- Designed an IoT platform optimising waste collection using real-time sensor data — demonstrating practical data analytics applied to urban infrastructure challenges.

TECHNICAL SKILLS

Monitoring, Evaluation & Data Collection:
KoboToolbox, Open Data Kit (ODK), SPSS/Stata, Microsoft Excel, Data Collection, Data Quality Assurance, Data Verification, Report Writing

Data Science & Analytics:
Python (Pandas, NumPy), Power BI, Data Visualization, Jupyter Notebook, Statistical Modeling, Agent-Based Modeling

Artificial Intelligence & Machine Learning:
TensorFlow, Convolutional Neural Networks (CNNs), Prompt Engineering, Large Language Model (LLM) Applications, Natural Language Processing (NLP)

Programming Languages:
Python, Java, JavaScript, C, C++

Web Development & APIs:
Django, FastAPI, REST APIs, Node.js, HTML5, CSS3

Database Management:
SQL, PostgreSQL, MySQL, SQLite, Database Design & Management

Networking:
Cisco Networking Fundamentals, TCP/IP, Network Configuration Basics

Blockchain Development:
Solidity, Ethereum, Smart Contracts, IPFS

DevOps & Development Tools:
Git, GitHub, Docker, Linux, Visual Studio Code (VS Code)

Professional Skills:
Data Analysis, Report Writing, Documentation, Problem Solving, Analytical Thinking, Leadership, Teamwork, Communication, Facilitation & Training, Mentorship, Adaptability, Time Management

LANGUAGES

English **Kiswahili** **Nuer** (Native)

CERTIFICATIONS & RECOGNITION

• Building LLM Applications with Prompt Engineering	*NVIDIA · July 2025*
• Data Science Certification	*Cisco Networking Academy · 2024*
• Microsoft Professional Learning Certifications & Badges	*Microsoft · Multiple*
• Prevention of Sexual Exploitation & Abuse (PSEA)	*UNHCR · May 2022*

• **Silver Academic Excellence Award — DAFI Conference, Mombasa**	*June 2024*
• Financial Literacy for Gig Workers	*Atingi · 2024*
• Basic Computer Training	*Don Bosco Technical Institute · 2024*
• Mathematics Contest Certificate	*SomaliBantu Secondary School · 2017*

INTERESTS & HOBBIES

- Coding and programming
- Reading
- Football
- Music and watching series

REFEREES

Dr. Lawrence Nderu

JHUB Africa Founder and Chairman of Computing Department

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

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